A Machine Learning Approach to Crop-Specific Irrigation Need Prediction in Nepal Using a Random Forest Classifier

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***Abstract*—Agriculture remains a major part of Nepal's economy, yet irrigation decisions in many areas are still reactive and not fully adapted to local soil and climatic conditions. This study presents a machine learning approach for predicting crop-specific irrigation need as a three-class problem: Low, Medium, and High. A synthetic Nepal-calibrated agricultural dataset from Kaggle was used, containing soil, climate, crop, and farm-management variables.**

The workflow included data validation, preprocessing, feature engineering, class imbalance analysis, and model evaluation. Three classification models were compared: Logistic Regression, Decision Tree, and Random Forest. SMOTE was also applied to examine whether minority-class handling improved performance. The models were evaluated using accuracy, macro precision, macro recall, macro F1-score, and confusion matrices.

The results showed that the baseline Random Forest model achieved the best overall performance, with a macro F1-score of 0.9678. SMOTE slightly improved the Decision Tree model but did not improve Random Forest or Logistic Regression. The findings indicate that Random Forest is effective for this irrigation classification task and can support future irrigation decision-support systems in Nepal.

I. Introduction

Agriculture continues to play a central role in Nepal's economy and rural livelihoods. The sector remains closely tied to household income, food security, and regional development, while productivity is strongly influenced by water availability, seasonal rainfall, and field-level management practices [1], [2]. In many parts of Nepal, irrigation decisions are still made using traditional schedules or visible signs of crop stress. Such reactive practices can lead to delayed intervention, inefficient water use, and reduced crop productivity [3].

These challenges have become more serious under increasing climatic variability. Irregular monsoon patterns, prolonged dry periods, and unseasonal rainfall have made irrigation planning more uncertain across Nepal's farming regions. The problem is further complicated by Nepal's geographical diversity, as agricultural conditions differ substantially between the Terai plains and the hill regions. As a result, irrigation demand cannot be managed effectively through a uniform rule-based approach, creating the need for data-driven methods that reflect local soil, climate, and crop conditions [4], [3].

Machine learning provides a suitable approach because it can model complex relationships among multiple agricultural variables and convert them into predictive decision-support tools. In this study, irrigation need was formulated as a three-class classification problem with output labels of Low, Medium, and High. A Nepal-calibrated agricultural dataset was used, containing variables related to soil characteristics, climatic conditions, crop type, irrigation practice, and regional context. Three classification models were compared: Logistic Regression, Decision Tree, and Random Forest. In addition, the effect of Synthetic Minority Over-sampling Technique (SMOTE) was examined because the High irrigation need class was the minority class in the dataset.

The study contributes a reproducible machine learning workflow for Nepal-focused irrigation classification. It emphasizes preprocessing, feature engineering, class imbalance handling, and final-model interpretation through confusion-matrix and feature-importance analysis.

II. Literature Review

Machine learning has increasingly been applied to agricultural decision-support tasks such as crop recommendation, irrigation automation, yield prediction, and soil health analysis. Existing studies indicate that these methods are particularly valuable when agricultural outcomes depend on interacting soil, weather, and management variables rather than on simple linear relationships [10], [9]. This is especially relevant to irrigation problems, where the required water level is influenced simultaneously by environmental conditions, crop development, and prior management actions.

Recent research further suggests that intelligent irrigation and crop-support systems can improve resource efficiency when environmental variables are combined with data-driven modelling. Earlier work has shown that adaptive irrigation systems, soil-informed recommendation models, explainable machine learning approaches, and engineered soil-related features can all strengthen agricultural prediction performance and improve trust in recommendation systems [5], [6], [7], [8].

At the methodological level, the present study is also informed by prior research on imbalanced classification and tabular supervised learning. SMOTE remains one of the most widely used approaches for minority-class oversampling [11], while broader work on learning from imbalanced data warns that resampling does not uniformly improve all classifiers and should be evaluated empirically rather than assumed to be

beneficial [12]. For structured tabular data, Random Forest continues to be one of the strongest and most interpretable ensemble methods [13], whereas Logistic Regression remains a useful linear baseline [14]. These studies together support the design choice of this paper: to compare multiple classifiers under identical preprocessing conditions and to evaluate the effect of SMOTE directly.

However, limited work is publicly documented on multi-class irrigation need prediction using a Nepal-calibrated dataset with explicit baseline-versus-SMOTE comparison. This study addresses that gap through a reproducible workflow combining preprocessing, feature engineering, imbalance handling, model comparison, and interpretation.

III. Problem Statement and Objectives

Nepal's agricultural sector remains highly dependent on seasonal rainfall and locally variable irrigation practices. In many farming areas, irrigation decisions are still made reactively, often after visible crop stress has already appeared. Such practices can result in both over-irrigation and under-irrigation, depending on local soil conditions, crop stage, and weather patterns. The absence of a simple data-driven prediction mechanism makes it difficult to estimate irrigation demand in advance, particularly across Nepal's diverse agricultural regions.

This study addresses that problem by formulating irrigation need prediction as a three-class classification task, where irrigation demand is categorized as Low, Medium, or High using soil, climate, crop, and management-related variables. The main objective of the study was to develop and evaluate machine learning models for this task and to identify the most effective model for predicting irrigation need. In addition, the effect of class imbalance handling through SMOTE was examined in order to determine whether resampling improved classification performance, particularly for the minority High irrigation need class.

IV. Dataset Description

The dataset used in this study was a synthetic agricultural dataset from kaggle which was calibrated to reflect Nepal's farming context. It contained a total of 900,000 records, divided into 630,000 training instances and 270,000 test instances. The training set contained 20 input features and one target variable, *Irrigation_Need*, while the test set contained only the input variables. The target variable was defined as a three-class label with the categories Low, Medium, and High.

The dataset included variables related to soil properties, climatic conditions, crop characteristics, irrigation practice, and regional context. These variables included soil type, soil pH, soil moisture, organic carbon, electrical conductivity, temperature, humidity, rainfall, sunlight hours, wind speed, crop type, crop growth stage, season, irrigation type, water source, field area, mulching use, previous irrigation amount, and region. The dataset was designed to reflect Nepal-specific agricultural patterns, including traditional seasons such as Barkhe, Hiunde, and Chaite, as well as the five historical development regions.

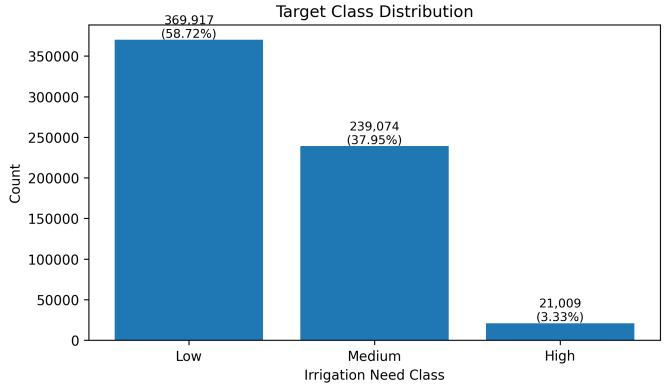

Fig. 1. Distribution of the target classes in the training dataset

Initial inspection showed that the dataset was clean and well-structured. No missing values or duplicate rows were found in either the training or test split. However, the target variable was imbalanced. In the training set, the Low class accounted for 58.72% of records, the Medium class accounted for 37.95% of records, and the High class accounted for only 3.33%. This imbalance was important because it increased the difficulty of correctly identifying the minority High irrigation need class during model training and evaluation. Figure 1 illustrates the class imbalance in the training data, where the Low and Medium classes are substantially more frequent than the High irrigation need class.

V. Methodology

A. Overview

A supervised machine learning workflow was followed to predict irrigation need as a three-class classification problem. The overall process consisted of data validation, preprocessing, feature engineering, model training, class imbalance handling, evaluation, and final model selection. Three classification models were used for comparison: Logistic Regression, Decision Tree, and Random Forest. These models were selected to represent baseline linear learning, interpretable tree-based learning, and ensemble tree-based learning.

B. Workflow Summary

The workflow followed this sequence:

$$\text{Raw Data} \rightarrow \text{Validation} \rightarrow \text{Preprocessing} \rightarrow \text{Feature Engineering} \rightarrow \text{Baseline Models} \rightarrow \text{SMOTE Models} \rightarrow \text{Evaluation} \rightarrow \text{Final Model}$$

This process ensured that data preparation, model training, imbalance handling, and evaluation were carried out in a consistent and reproducible manner.

C. Machine Learning Models

Three classification models were selected for comparison: Logistic Regression, Decision Tree, and Random Forest.

Logistic Regression was used as a simple linear baseline. Decision Tree was included because it can capture non-linear patterns and provide interpretable rule-based predictions. Random Forest was selected as the final ensemble model because it combines multiple decision trees, improves stability, performs well on structured tabular data, and provides feature-importance scores for interpretation.

D. Data Preprocessing

The training and test files were first checked for structural consistency, duplicate rows, and missing values. Since no missing values or duplicate records were found, no imputation or record removal was required. The identifier column was removed before modeling because it had no predictive value. The dataset was then divided into training and validation subsets using a stratified 80:20 split so that the target-class distribution was preserved.

Categorical variables were transformed through encoding, while numerical variables were standardized within the preprocessing pipeline. This step was particularly important for Logistic Regression and the SMOTE-based setting, where distance- and scale-sensitive operations can be affected by inconsistent feature magnitudes. For the SMOTE experiment, preprocessing was adapted so that mixed numerical and categorical variables could be handled appropriately within the resampling workflow. Particular attention was given to keeping preprocessing consistent across models so that differences in performance could be attributed more confidently to the learning algorithms themselves rather than to differences in data handling.

E. Feature Engineering

Feature engineering was introduced to capture interactions among the original variables and to reflect irrigation-related relationships more explicitly. Four additional features were created:

$$\text{Moisture Deficit} = \text{Humidity} - \text{Soil Moisture} \tag{1}$$

$$\text{Soil Stress Score} = |\text{Soil pH} - 7| + \text{Electrical Conductivity} \tag{2}$$

$$\text{Temperature–Rainfall Ratio} = \frac{\text{Temperature_C}}{\text{Rainfall_mm} + 1} \tag{3}$$

$$\text{Irrigation per Hectare} = \frac{\text{Previous Irrigation_mm}}{\text{Field Area_hectare} + 1} \tag{4}$$

These features were intended to represent water deficit, soil-related irrigation stress, the relationship between atmospheric demand and rainfall input, and irrigation intensity relative to field size. The inclusion of engineered features helped strengthen the practical relevance of the model by embedding domain-informed relationships into the feature space.

F. Model Training and Imbalance Handling

Three classifiers were trained and evaluated. Logistic Regression was used as a linear baseline with a maximum iteration limit of 1000 and class weighting [14]. Decision Tree was configured with controlled depth and minimum split constraints in order to reduce overfitting, while Random Forest was trained as an ensemble model using 200 trees [13]. In the baseline setting, class-weight adjustment was used where appropriate to reduce bias toward the majority class. In the second setting, SMOTE-based resampling was applied before model training in order to examine whether minority-class representation could be improved [11], [12].

G. Evaluation Metrics

Model performance was assessed using accuracy, macro-averaged precision, macro-averaged recall, macro-averaged F1-score, and confusion matrices. Macro F1-score was treated as the main comparison metric because it gives equal importance to all three classes, including the minority High irrigation need class. The final model was selected on the basis of overall predictive performance and class balance rather than accuracy alone.

VI. Implementation and Reproducibility

The project was implemented in Python using both Jupyter Notebook and modular Python scripts. Pandas and NumPy were used for data handling, scikit-learn and imbalanced-learn for model development, and Matplotlib and Seaborn for visualization [15].

The workflow generated model summaries, classification reports, confusion matrices, feature-importance outputs, the final Random Forest model, and test predictions. This structure supports reproducibility and allows the experiments to be rerun from the submitted code and instructions.

VII. Results and Discussion

The comparative results showed that Random Forest achieved the strongest overall performance among all evaluated models. In the baseline setting, Random Forest produced an accuracy of 0.9850 and a macro F1-score of 0.9678, which was the highest among the three models. Decision Tree also performed well, with a macro F1-score of 0.9523, while Logistic Regression produced a considerably lower macro F1-score of 0.7435. Table I summarizes the performance of all models under both the baseline and SMOTE-based settings. These results indicate that tree-based models were more suitable for this classification task than the linear baseline model.

All precision, recall, and F1 values reported in Table I are macro-averaged metrics. The effect of SMOTE was then examined to determine whether minority-class handling improved model performance. The results showed that SMOTE did not improve all models equally. A small improvement was observed for Decision Tree, where the macro F1-score increased from 0.9523 to 0.9553. However, the performance of Random Forest decreased slightly after SMOTE, with macro F1-score falling from 0.9678 to 0.9637. Logistic Regression

TABLE I
VALIDATION PERFORMANCE OF BASELINE AND SMOTE-BASED MODELS

Setting	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Baseline	Logistic Regression	0.8430	0.7059	0.8509	0.7435
Baseline	Decision Tree	0.9815	0.9394	0.9671	0.9523
Baseline	Random Forest	0.9850	0.9680	0.9677	0.9678
SMOTE	Logistic Regression	0.7087	0.5893	0.7401	0.6224
SMOTE	Decision Tree	0.9824	0.9476	0.9638	0.9553
SMOTE	Random Forest	0.9833	0.9619	0.9656	0.9637

also performed worse after SMOTE, with its macro F1-score decreasing from 0.7435 to 0.6224. These findings suggest that although resampling increased representation of the minority class, it did not necessarily lead to better overall generalization.

A closer inspection of the class-wise results showed that Logistic Regression achieved relatively high recall for the High irrigation need class, but this came at the cost of weaker balance across the full classification task. In contrast, Random Forest maintained strong and stable performance across all three classes, which explains its superior macro F1-score. Since macro F1-score gives equal importance to each class, including the minority class, it was treated as the main selection criterion. On this basis, the baseline Random Forest model was selected as the final model.

Overall, the results indicate that irrigation need prediction in this dataset was better captured by nonlinear tree-based learning than by linear classification. The strong performance of Random Forest also suggests that the interaction between soil, crop, climate, and irrigation-related variables was complex and not easily represented by a simple linear boundary. While SMOTE was useful as a comparative imbalance-handling method, the final findings showed that the best-performing model remained the baseline Random Forest trained on the original dataset.

A. Interpretation of the Final Model

Figure 2 shows the class-wise prediction performance of the final baseline Random Forest model. Strong classification performance was observed across all three irrigation levels, with particularly high consistency for the Low and Medium classes. The model also performed well on the minority High irrigation need class, which supports its selection as the final model.

To better understand the behaviour of the final selected model, feature-importance analysis was carried out using the fitted baseline Random Forest classifier. Table II presents the five most influential predictors identified by the model. The results showed that Soil_Moisture was the most important feature, followed by Temperature_C, Wind_Speed_kmh, Rainfall_mm, and Crop_Growth_Stage_Harvest. These findings are consistent with the agricultural logic of irrigation demand, since water requirement is strongly influenced by existing soil water availability, atmospheric conditions, and crop development stage.

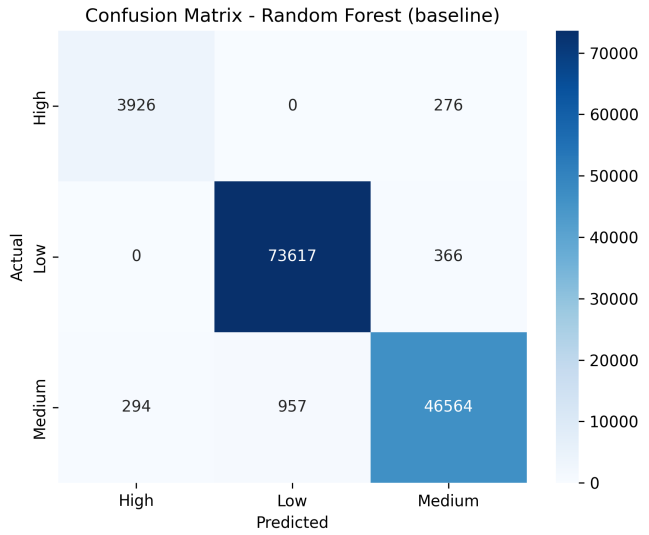

Fig. 2. Confusion matrix for the baseline Random Forest model on the validation set

TABLE II
TOP FIVE FEATURE IMPORTANCES FROM THE FINAL RANDOM FOREST MODEL

Feature	Importance
Soil_Moisture	0.2406
Temperature_C	0.1254
Wind_Speed_kmh	0.0963
Rainfall_mm	0.0706
Crop_Growth_Stage_Harvest	0.0690

The strong performance of Random Forest suggests that irrigation need in this dataset was determined by nonlinear interactions among environmental and management-related variables rather than by simple linear relationships. In particular, the combined effect of low soil moisture, reduced rainfall, higher temperature, and stronger drying conditions likely contributed to stronger separation of the High irrigation need class. This also helps explain why Random Forest outperformed Logistic Regression, as the latter is less capable of capturing complex feature interactions.

B. Practical Implications

The findings suggest that irrigation prediction should consider multiple factors together, including soil condition, recent irrigation history, climate, and crop stage. A model of this kind could support irrigation scheduling, assist extension workers, and contribute to future decision-support tools for precision agriculture in Nepal.

VIII. ETHICAL, LEGAL, AND PROFESSIONAL CONSIDERATIONS

Although the final model achieved strong performance, it should be used responsibly. The dataset was synthetic and downloaded from Kaggle, so it may not fully represent real farming conditions in Nepal. Without validation on field data,

incorrect predictions could lead to under-irrigation, reduced yield, over-irrigation, or water waste.

Professionally, the model should be treated as a decision-support tool rather than a replacement for farmer judgment or agricultural expertise. Any real deployment should consider data quality, transparency, local validation, and fair access for smallholder farmers.

IX. Limitations

Several limitations should be acknowledged. First, the dataset was synthetic and was downloaded from Kaggle rather than collected directly from field observations. Although it was designed to reflect Nepal's agricultural context, it may not fully capture real-world variability, measurement noise, and local farming conditions.

Second, crop-region patterns and field-size distributions may not fully match actual smallholder farming patterns in Nepal. This may affect generalisability, especially where irrigation behaviour depends on local resource constraints.

Finally, the dataset had no temporal structure, so seasonal irrigation changes could not be modelled directly. Economic factors such as pumping cost, fuel use, and water availability were also not included, even though they may influence real irrigation decisions.

X. Conclusion

This study presented a machine learning approach for predicting irrigation need in Nepal as a three-class classification problem. Three models, namely Logistic Regression, Decision Tree, and Random Forest, were developed and evaluated using a structured preprocessing pipeline and two experimental settings: baseline training and SMOTE-based imbalance handling. The results showed that Random Forest achieved the strongest overall performance, with the highest macro F1-score among all evaluated models.

The comparative analysis also showed that SMOTE did not improve all models equally. A slight improvement was observed for Decision Tree, whereas the performance of Random Forest and Logistic Regression decreased after resampling. Based on the overall results, the baseline Random Forest model was selected as the final model. The study contributes a reproducible machine learning workflow for Nepal-specific irrigation need prediction, a direct baseline-versus-SMOTE comparison, and an interpretable final Random Forest model supported by confusion-matrix and feature-importance analysis. These findings indicate that tree-based ensemble learning was the most effective approach for this irrigation classification task and provided the best balance across all three classes.

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Appendix

GitHub Repository: nepal_irrigation_ml

Video link: nepal_irrigation_ml_demo